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Smoke vehicle detection based on robust codebook model and robust volume local binary count patterns☆,☆☆

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ABSTRACT

Smoke vehicle detection is still vulnerable to false alarms. To reduce false alarm rates, this paper presents a smoke vehicle detection method based on Robust Codebook (R-Codebook) model and Robust Volume Local Binary Count (R-VLBC) pattern. In detecting suspected smoke blocks, we propose the R-Codebook model, which is robust to illumination changes and camera shaking or leaf shaking in the scene. In recognizing smoke block sequences, firstly, we propose a robust and discriminative descriptor called Non-Redundant VLBC (NR-VLBC). The feature dimension is reduced by half than the VLBC to overcome the issue that the VLBC is sensitive to the relative changes between background and foreground. Secondly, we propose a new Completed VLBC (CVLBC) by combining VLBC-Sign, VLBC-Magnitude, VLBC-Center gray level and VLBC-Difference to characterize spatial–temporal features. Thirdly, two strategies are used to further improve the proposed CVLBC to form R-VLBC. One is to overcome the issue that the CVLBC is sensitive to noise based on using a weighted local threshold (WLT), which is robust to noise and illumination variants and also make a balance between noise resistance and information of individual pixel. The other one is to extract multi-scale information by using a set of radius and sampling points. Extensive experiments show that the proposed method can achieve better performances than existing methods.

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1. Introduction

The vehicle belching visible black smoke from the vehicle exhaust pipe is usually called a smoke vehicle. The smoke vehicle seriously pollutes the air and endangers human health. Fig. 1 shows a typical example of smoke vehicles. We can see heavy smoke at the back of the vehicle. This paper is focused on how to automatically detect smoke vehicle from traffic surveillance videos.

In the real world application, we mainly detect smoke vehicles by inspecting on the road or watching surveillance videos in the monitoring room. There is no doubt that these methods are ineffective. Currently we want to detect smoke vehicles based on computer vision technology to make it more effective and intelligent.

The core of vision-based smoke vehicle detection is smoke detection. Although lots of literatures in smoke detection can be transplanted to smoke vehicle detection, it is still vulnerable to false alarms in complex situations, due to the variability of shape, motion and irregular patterns of the smoke behind smoke vehicles, as well as the interference from illumination, camera noise and shadows.

To solve these issues and reduce false alarms, we propose a smoke vehicle detection method based on Robust Codebook (R-Codebook) model and Robust Volume Local Binary Count (R-VLBC) pattern. The main contributions include:

(1) We propose the R-Codebook model to detect suspected smoke blocks. The proposed R-Codebook does not need to construct codebook for every pixel, and so obtains a faster speed than original Codebook model. The proposed R-Codebook model utilizes the light insensitivity of LBP to enhance the adaptability to light changes, and is robust to illumination changes in the scene. In addition, it is robust to camera shaking and leaf shaking, which are commonly occurred in the traffic surveillance videos.
(2) We propose the Non-Redundant VLBC (NR-VLBC). Compared with the original VLBC, the proposed NR-VLBC is more robust and discriminative to reflect the relative contrast between the background and foreground. In addition, the original VLBC code

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Fig. 1. A smoke vehicle with visible black smoke from the vehicle exhaust pipe. The region marked by the red circle is the position of smoke. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and its complement are considered as the same case. Thus the dimension of feature vector will be reduced by half.

(3) We propose a new Completed VLBC (CVLBC) by combining the VLBC_S, VLBC_M, VLBC_C and VLBC_D to characterize spatiotemporal features. The VLBC_S is equivalent to the above proposed NR-VLBC. The VLBC_M measures the local variance of magnitude. The VLBC_C expresses the image local gray level. The VLBC_D expresses the gradient information of neighborhood pixels.

(4) Two strategies are used to further improve the proposed CVLBC to form R-VLBC. Firstly, the original VLBC is sensitive to noise since it only use the central pixel as a threshold directly. We use weighted local threshold (WLT) as a threshold instead of the intensity value of the center pixel. Compared to the center pixel, WLT is more robust to noise and illumination variants. A parameter is also introduced to make a balance between noise resistance and information of individual pixel. Secondly, because the smoke with different distances from the camera will present different scales, and so we introduce a multi-scale strategy by using a set of radius and sampling points to extract multi-scale information to recognize smoke sequences.

The rest of this paper is organized as follows: Section 2 introduces the related works. Section 3 describes the detection of suspected smoke blocks. In Section 4, we describe the proposed R-VLBC descriptor in details. The experimental results are demonstrated in Section 5, followed by conclusion drawn in Section 6.

2. Related work

To date, there are lots of literatures in vision-based smoke detection. From the view of the used features, we roughly divide them into three categories, as shown in Table 1.

2.1. Spectral features [1–8]

The basic spectral features are color features, which are widely used in smoke detection since the color of the smoke is always black or cream white. Color features are discriminative but they have several limitations in the presence of \illumination changes, camouflage and shadows. Therefore, most existing methods always fuse the color

Table 1
The used features in smoke detection literatures.

Feature kinds		Features and literatures
Spectral features		Image graininess, intensity values and histogram [1], Mean, variance, skewness and kurtosis [2,43], Smoke color filtering [3], chromatic configuration [43] Color moment (CM) [5], Assessing smoke probability [6], Intensity and saturation density in the HSI color space [7], rule-based chrominance detection [28] Rule and threshold based color features [4,8,17,18,39–42],
Spatial features	Texture features,	Local binary patterns (LBP) [7,14,15,34,47], LBP pyramids, LBP variance (LBPV) pyramids [9], Multi-scale block Tamura (MSB-Tamura) [10], Pairwise comparing LBP (PCLBP) [13], Non-redundant LBP (NRLBP) [16]. Variable Orientation GLCM (V-GLCM) [3] High-order Local Ternary Patterns (HLTP) [12] Hamming-distance-based LBP (HDLBP) [46]; LBP co-occurrence with HDLBP (LBPCoHDLBP) [46] Gray level co-occurrence matrix (GLCM) [11,20,32,34], HLTP based on magnitudes of noise removed derivatives and values of center pixels (HLTPMC) [12];
	Gradient-based features	Shape features [3,6,17], Contour [2], Edge orientation histogram (EOH) [7], Hu moments [2,7]. Histogram of oriented gradient (HOG) [14,18,43]; Haar-like [7,19]; Range filtering on three orthogonal planes (RF-TOP) [5]; Surface roughness and area randomness [41]
	Spatial learning-based features	Dictionary learning [22,23]. Convolutional neural networks (CNN) [24,27]; Deep multi-scale CNN (DMCNN) [25]; Deep normalization and CNN (DNCNN) [26] Holistic learning-based high-order feature [48]
	Spatial frequency domain features	Discrete wavelet transform (DWT) and wavelet [4,20,21,28,34,43], Discrete cosine transform (DCT) [21,22].
Temporal/ spatiotemporal features	Motion and optical flow	Histogram of oriented optical flow (HOOF) [18], Spatiotemporal bag-of-features [18] Motion orientation histogram (MOH) [5], Motion orientation model [40], Optical smoke flow pattern using Gabor filter [8], motion vector [41,43] Optical flow [42]; transmission [44], Trajectories, right-leaning line, smooth streamline, low-frequency, source and vertical–horizontal ratio [31] Motion energy and saliency map [45]. Sharp edge; growing region; rising region detection, perimeter disorder [39]
	Dynamic texture	Volume LBP (VLBP) [29], LBP on three orthogonal planes (LBPTOP) [29], Complete VLBP (CVLBP) [29], swaying and diffusion model based on GLCM [32], Non-redundant local motion binary patterns (NRLMBP) [16] Map of brightness differences, edge map, and Laplacian map in Spatio-Temporal LBP (STLBP) [30] Weber local descriptors in three orthogonal planes (WLD-TOP) [33],
	Spatiotemporal math model	Surfacelet transform and HMT model [34,49], higher order linear dynamical systems (hLDS) [35],
	Spatiotemporal learning-based features	Spatial–temporal ConvNet [36], recurrent convolutional network [37], Multi-scale and multi-order features from 3D local differences [38].
	Spatiotemporal frequency domain features	Spatial–temporal wavelet [33], spatial–temporal energy using DWT [8]

features with other kinds of features to recognize the smoke-colored objects, such as black road surface, black shadows and black vehicle body.

2.2. Spatial features [9–28,46–48]

The spatial features include texture features, gradient-based features, spatial learning-based features and spatial frequency domain features, etc. Different kinds of features have different properties that allow us to handle the critical situations differently (illumination changes, motion changes, structure background changes). For example, edge features handle the local illumination changes and the ghost left when waking foreground objects begin to move. In recent years, the related topics of smoke detection in computer vision community have ranged from spatial features to spatiotemporal features. The smoke-liked objects easily lead to false alarms due to the absence of spatiotemporal features.

2.3. Temporal/spatiotemporal features [29–45]

Lots of temporal features and spatiotemporal features have been studied. The temporal features can capture the dynamic characteristics of the smoke and so always achieve better performances. Existing motion and optical flow features used in smoke detection can characterize the temporal features, but without abundant spatial features. The dynamic texture descriptor can characterize spatiotemporal features, such as VLBP. But for an example with radius 1 and sampling points 8, the feature dimension of VLBP will be high to $2^{3\times8+2} = 67,108,864$, and this makes it useless in smoke detection.

Therefore, although various features have been investigated, it is still vulnerable to false alarms in some complex situations, due to the variability of shape, motion and irregular patterns of the smoke in smoke vehicle detection, as well as the interference from illumination, camera noise and shadows.

3. Detection of suspected smoke blocks

In this section, a new R-Codebook model and a block-based strategy is introduced to detect suspected smoke blocks.

3.1. Moving object detection

The smoke vehicle is usually seen as moving objects in the traffic surveillance videos. This can be utilized to distinguish a true smoke from other smoke-like background objects, such as the smoke-colored road surface regions. To eliminate the disturbance of these non-smoke objects and also reduce the computational cost of subsequent feature extraction, we first detect the moving object by using a new Robust Codebook (R-CodeBook) model.

The original Codebook model [50] was proposed by Kim and has been improved by many researchers. The core of this model is that the sample background values at each pixel are quantized into codebooks which represent a compressed form of background model for a long image sequence. The Codebook model can capture structural background variation due to periodic-like motion over a long period of time under limited memory. It can handle scenes containing moving backgrounds or illumination variations to robust detection.

The original Codebook model [50] is summarized as follows:

(1) Construction of the initial codebook.

Let $C = \{c_1, c_2, \ldots, c_L\}$ denote the Codebook of a pixel with L codewords. Each codeword c_i contains an RGB vector v_i and a 6-tuple aux_i, i.e.,

$$v_i = \left(\overline{R}_i, \overline{G}_i, \overline{B}_i\right) \tag{1}$$

$$aux_i = \left\langle I_i^{(\min)}, I_i^{(\max)}, f_i, \lambda_i, p_i, q_i \right\rangle \tag{2}$$

where $v_i = (\overline{R}_i, \overline{G}_i, \overline{B}_i)$ is the RGB intensity values. $I_i^{(\min)}$ and $I_i^{(\max)}$ are the min and max brightness, respectively. f_i is the occurring frequency of current codeword. λ_i is the maximum negative run-length (MNRL). p_i and q_i are the first and last access times, respectively.

(2) Codebook model and updating.

Let $X = \{x_1, x_2, \ldots, x_N\}$ be a training sequence for a single pixel consisting of N RGB-vectors. Let x_t be the value sampled at time t. To find the best codeword it matches, the color distortion measure and the brightness bounds are employed.

$$\text{Color distortion}: \quad \sqrt{\|x_t\|^2 - \langle x_t, v_m\rangle^2 / \|v_m\|^2} \le \varepsilon_1 \tag{3}$$

$$\text{Brightness bounds}: \quad I_{low} \le \|x_t\| \le I_{high} \tag{4}$$

where I_{low} and I_{high} are the low bound and high brightness bound, respectively. ε_1 is a sampling threshold. v_m is the matched RGB vector.

If codebook C is empty or there is no match, we create a new codeword c_L by setting

$$L \leftarrow L+1 \quad aux_L \leftarrow (I, I, 1, t-1, t, t); \quad v_L \leftarrow x_t \tag{5}$$

Otherwise, update the matched codeword c_m, by setting,

$$v_i \leftarrow \left(\frac{f_m\overline{R}_m + R}{f_m+1}, \frac{f_m\overline{G}_m + G}{f_m+1}, \frac{f_m\overline{B}_m + B}{f_m+1}\right) \tag{6}$$

$$aux_i \leftarrow \left\langle \min\left\{I, I_m^{(\min)}\right\}, \max\left\{I, I_m^{(\max)}\right\}, f_m+1, \max\{\lambda_m, t-q_m\}, p_m, t\right\rangle \tag{7}$$

For each codeword c_i, wrap around λ_i by setting

$$\lambda_i = \max\{\lambda_i, (N - q_i + p_i - 1)\} \tag{8}$$

Find the maximum time interval that the codeword has not recurred during the training period, and eliminate them,

$$M = \{c_m | c_m \in C, \lambda_m \le T_M\} \tag{9}$$

where T_m is set to half of the total frame number.

(3) Moving object detection.

We compute the distance of the sample from the nearest cluster mean. A detection threshold ε_2 is used. The pixel is detected as foreground if no acceptable matching codeword exists. Otherwise it is classified as background.

The original Codebook model has achieved great success. However, the model is still with a slow speed in complex environment. In order to achieve a real-time and robust detection of video moving objects, a new Robust Codebook (R-Codebook) is proposed. Our improvements include: 1) A strategy is used to speed up algorithm; 2) The local binary patterns (LBP) is added to characterize the local texture features and improve algorithm robustness to illumination changes in the scene.

To better understand the proposed idea, we first make a brief introduction of the LBP code. The original LBP code was proposed by Ojala [51] and used for texture classification. It can reflect the relation of intensities between a center pixel and its neighbors.

The original LBP code [51] with radius R and sampling points' number P is computed as follows:

$$LBP_{P,R} = \sum_{i=0}^{P-1} s(g_i(R) - g_c)2^i \tag{10}$$

$$s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{11}$$

where g_c denotes the intensity value of the center pixel, $g_i(R)$ denotes the i-th sampling value of the circular neighborhood with radius R.

Ojala [50] also proposed the uniform, rotation invariant, and rotation invariant uniform patterns, respectively.

$$U(LBP_{P,R}) = \sum_{i=0}^{P-1} \left| s\left(g_{\mathrm{mod}(i+1,P)}(R) - g_c\right) - s(g_i(R) - g_c) \right| \tag{12}$$

$$LBP_{P,R}^{RI} = \min_{0 \leq i \leq P-1} \{ROR(LBP_{P,R}, i)\} \tag{13}$$

$$LBP_{P,R}^{RIU2} = \begin{cases} LBP_{P,R}^{RI}, & \text{if } U(LBP_{P,R}) \leq 2 \\ P+1, & \text{else} \end{cases} \tag{14}$$

where $U(LBP_{P,R})$ denotes the uniform pattern code. $LBP_{P,R}^{RI}$ denotes the rotation invariant pattern. $LBP_{P,R}^{RIU2}$ denotes the rotation invariant uniform pattern. mod(a, b) denotes the modulo operation. $ROR(x,i)$ means a circular bit-wise right shift for i times on the P-bit number x.

If P is set to 8, the histogram dimension is 256. It can be reduced to 58, 35 or 9 by applying uniform (U2), rotation invariant (RI) or rotation invariant and uniform (RIU2) mapping patterns, respectively.

The main idea of the proposed R-Codebook model is replacing the original RGB vector to the LBP histogram vector to record the texture features of local region using codebook. Firstly, we divide the frame image into blocks of size $N_b \times N_b$, and model each block by a codebook. Each codeword c_i in the codebook contains an LBP histogram vector v_i and a 6-tuple aux_i. $X = \{x_1, x_2, \ldots, x_N\}$ is a training sequence for a single pixel consisting of N LBP feature vectors. x_i is the value sampled at time t. In addition, we also introduce a weight parameter W to replace parameters λ_i, p_i, q_i.

In the proposed method, Eqs. (1), (2) are changed by

$$v_i = (v_i^1, v_i^2) \tag{15}$$

$$aux_i = \langle v_i, f_i, W_i \rangle \tag{16}$$

v_i^1 is the maximal value of the 9 elements in the histogram feature vector of RIU2 mapping patterns. v_i^2 is the texture type of v_i^1.

Based on the changes of codebook, Eqs. (5)–(7) are changed by

$$x_t = v_m \tag{17}$$

$$histSim(x_t, v_m) \geq \varepsilon_1 \tag{18}$$

where $histSim(x_t, v_m)$ denotes the vector similarity between x_t and v_m using histogram intersection method.

The Eqs. (5)–(7) are changed by

$$L \leftarrow L + 1; \quad aux_L \leftarrow (x_t, 1, t-1, t, t); \quad v_L \leftarrow x_t \tag{19}$$

$$v_i \leftarrow x_t \tag{20}$$

$$aux_i \leftarrow \langle v_m, \ f_m + 1, \ \min\{W_i + 1, 20\}, t \rangle \tag{21}$$

We check the access frequency of a codeword regularly. If it is larger than a fixed threshold, the corresponding pixel can be seen as a real background pixel during this period, then we can increase its weight. Otherwise, its weight should be reduced.

In the codeword matching, it should be noted that we just find the first codeword to satisfy Eqs. (17), (18). In addition, the most recently updated codeword is relocated to the front of the codebook list to speed up the algorithm.

The proposed R-Codebook does not need to construct codebook for every pixel, and so obtains a faster speed than original Codebook model. The R-Codebook model utilizes the light insensitivity of LBP to enhance the adaptability to light changes, and is robust to illumination changes in the scene. In addition, it is robust to camera shaking and leaf shaking, which are commonly occurred in the traffic surveillance videos.

3.2. Suspected smoke blocks

After detecting the moving objects, we will partition the whole frame and take the moving-object-covered blocks as suspected smoke blocks. Fig. 2 shows a results of the suspected smoke blocks (yellow blocks) in one frame using R-Codebook model.

After detecting a suspected smoke block, we will extract its front blocks and back blocks in the same position of different frames to form a block image sequence and make further analysis to classify the sequence to smoke or non-smoke.

4. Robust volume local binary count (R-VLBC)

In this section, the proposed robust volume local binary count (R-VLBC) is introduced.

From Table 1, we can see that lots of temporal/spatiotemporal features have been introduced into smoke (vehicle) detection. The optical flow based features mainly focus on the temporal characteristics, but lack of spatial characteristics. The dynamic textures descriptor VLBP [29] extracts histograms of thresholded local spatiotemporal volumes using both appearance and motion features, but with high feature dimension.

The R-VLBC is mainly proposed to solve the following four issues of VLBC. (1) The original VLBC is sensitive to the relative changes between the background and foreground. (2) The original VLBC is sensitive to noise since it only use the central pixel as a threshold directly. It is needed to make a balance between noise resistance and information of individual pixel. (3) The VLBC_D is introduced to expresses the gradient information of neighborhood pixels. (4) The smoke with different distances from the camera will present different scales, but the VLBC cannot express the multi-scale information.

4.1. The original local binary count (LBC)

The local binary count (LBC) [52] was used for 2D gray-scale texture classification with invariance to rotation. The encoding scheme of LBC is given by,

$$LBC_{P,R} = \sum_{i=0}^{P-1} s(g_i(R) - g_c) \tag{22}$$

where g_c denotes the intensity value of the center pixel, $g_i(R)$ denotes the ith pixel, which are equally spaced on a circle of radius R. $s(x)$ can be seen in Eq. (11).

The LBP descriptor focuses on local structure features based on different patterns, but LBC only considers how many pixels have a relatively higher intensity value than the central pixel in the local neighborhood. LBC can characterize the macroscopic texture structures other than the micro-structures extracted by LBP. Because LBC does not involve any local structure information, it is naturally rotation invariant, i.e., if we rotate the local neighborhood, the LBC code remains the same, but the LBP code may changes. LBC is discriminative and robust to illumination changes. In addition, it is efficient in storage requirement and computational complexity.

4.2. Volume LBC (VLBC)

To combine appearance and motion information together for block sequences recognition, the 2D texture LBC is extended to spatiotemporal domain to form 3D texture volume LBC (VLBC) [53].

Let $V^{(3P+3)}$ denote a $3P+3$ pixels set of a sub-volume of three block images in three sequential frames with their positions being the same in each frame. There can be more frames in the sub-volume. $V^{(3P+3)}$ is a collection of pixels and given by,

$$V^{(3P+3)} = \{g_{t_c-L,c}, g_{t_c-L,0}, \cdots, g_{t_c-L,P-1}, g_{t_c,c}, g_{t_c,0}, \cdots, g_{t_c,P-1} \; g_{t_c+L,0}, \cdots, g_{t_c+L,P-1}, g_{t_c+L,c}\} \tag{32}$$

where $g_{t_c-L,\,c}$ and $g_{t_c+L,\,c}$ are the intensity values of central pixels in the first and third block with time interval to central block, respectively. $g_{t,\,p}$ $(t = t_c - L, t_c, t_c + L; p = 0, 1, \ldots, P-1)$ are the equally spaced P neighboring points on a circle of radius R in patch t. $g_{t_c,\,c}$ is the intensity value of the central pixel in $V^{(3P+3)}$.

We remove the term $g_{t_c,\,c}$ in Eq. (32) and leave $3P+2$ items in $V^{(3P+3)}$. We represent the set as

$$V^{(3P+2)} = \{v_0, v_1, \ldots, v_{3P+1}\} \tag{24}$$

The central pixel is used as a threshold of the sample pixels to compute the difference information,

$$\begin{aligned} V^{(t)} &= \{g_{t_c-L,c} - g_{t_c,c}, g_{t_c-L,0} - g_{t_c,c}, \cdots, g_{t_c-L,P-1} - g_{t_c,c}, \\ &\quad g_{t_c,0} - g_{t_c,c}, \cdots, g_{t_c,P-1} - g_{t_c,c} \\ &\quad g_{t_c+L,0} - g_{t_c,c}, \cdots, g_{t_c+L,P-1} - g_{t_c,c}, g_{t_c+L,c} - g_{t_c,c}\} \\ &= \{v_0 - g_{t_c,c}, v_1 - g_{t_c,c}, \ldots, v_{3P+1} - g_{t_c,c}\} \end{aligned} \tag{25}$$

where the term $g_{t_c,\,c} - g_{t_c,\,c}$ is omitted.

To model the distribution of difference and achieve invariance to scaling of the gray value, we keep only the sign of each term and count the number of positive signs. The VLBC code can depict the macroscopic structure of the local volume. All the differences between the central pixel and its neighboring pixels in a local volume are recorded in $VLBC_{L,\,P,\,R}$,

$$VLBC_{L,P,R} = \sum_{p=0}^{3P+1} s(v_p - g_{t_c,c}) \tag{26}$$

where p is the sequential index of elements in $V^{(3P+2)}$. $s(x)$ is defined in Eq. (11).

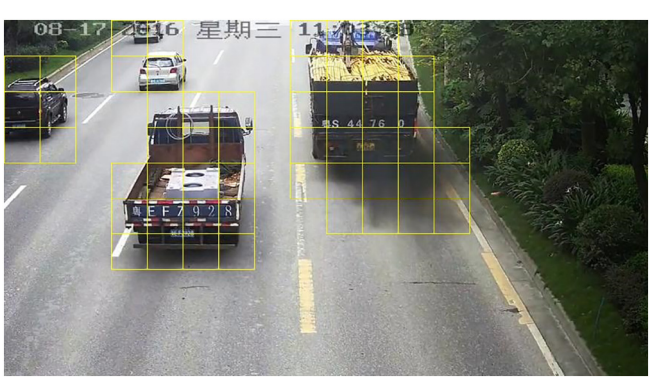

Fig. 2. Suspected smoke blocks (yellow blocks) using R-Codebook model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 3 is the entire process to compute $VLBC_{1,\,8,\,1}$. We sample the neighboring pixels (black point) in V, and then the sample points are thresholded with the central pixel (red point) in the central patch. Finally, we count the number of 1 in the thresholded values to generate the VLBC code to characterize the appearance and motion information of the volume.

4.3. The proposed robust VLBC (R-VLBC)

To further improve the robustness of VLBC, we propose the Robust VLBC (R-VLBC) based on four proposed strategies. The four proposed strategies used to improve the VLBC are the main contributions.

4.3.1. Ant-noise features

For convenience, we first introduce our ideas on the 2D texture descriptor LBC. It is easy to be extended to 3D texture descriptor VLBC.

Although the original LBC is powerful, it is sensitive to noise. As shown in Fig.4, the two patterns have the same LBC code, but their local structures are obviously different to each other. Therefore, the LBC is sensitive to noise since it only use the central pixel as a threshold directly.

To find a threshold which is insensitive to noise and invariant to monotonic gray scale transformation, we introduce the weighted local threshold (WLT), i.e., the value of each center pixel in a local area is replaced by WLT. The WLT is defined as,

$$WLT = \frac{\alpha \sum_{p=0}^{3P+1} v_p + g_{t_c,c}}{\alpha(3P+2)+1} \tag{27}$$

where v_p and $g_{t_c,\,c}$ are defined in Eq. (24). α is a parameter set by user. The WLT is equivalent to the conventional average local gray if α is set to 1, and the WLT is equivalent to the original LBC if α is set to 0.

Then the LBC process is applied by using WLT as a threshold instead of the intensity value of the center pixel. Obviously, compared to the center pixel, WLT is more robust to noise and illumination variants. The parameter α is used to make a balance between noise resistance and the information of individual pixel.

4.3.2. Non-redundant features

For convenience, we first introduce our ideas on the 2D texture descriptor LBC. It is easy to be extended to 3D texture descriptor VLBC.

Although the original LBC is powerful, it is sensitive to the relative changes between the background and foreground. As shown in Fig. 5, the LBC codes of the red rectangular regions in these two images are quite different from each other (α is set to 0). However, the two images (I_1 and I_2) describe the same spatial characteristics of smoke, since $I_1 + I_2 = 255$. In addition, the original LBC may has high requirement about computation and storage in some cases. For real-time smoke detection, we should lower the requirement of computation and storage as much as possible.

Taking the above two considerations, we propose the Non-Redundant LBC (NRLBC) to make it insensitive to the relative changes between the background and foreground and capture the appearance information of smoke.

A sample idea can be obtained based on the fact that the LBC codes in the two images are complementary to each other (the sum of them is P). However, if there are one or more neighborhood pixels equaling to the center pixel, the complementary characteristics are not satisfied. We noted that the above case often occurs, since the smoke has a smoothing effecting, which makes the center pixel equal to one or more of its neighborhood pixels. As shown in Fig. 5, the LBC codes of the two complementary images are not complementary, i.e., the sum of them is not 8.

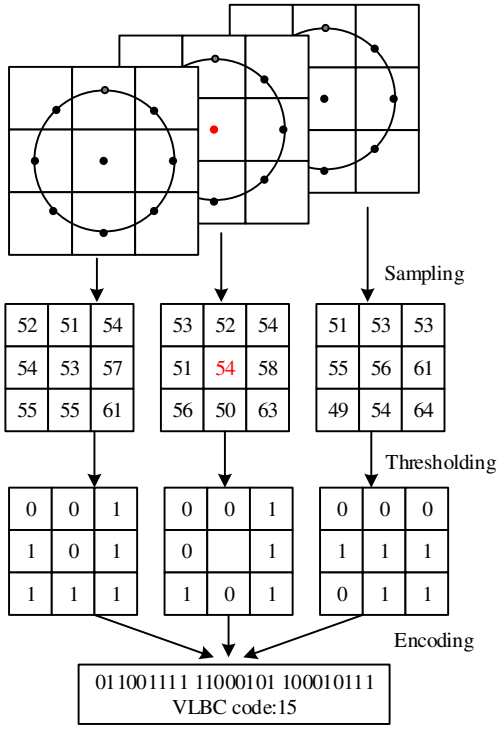

Fig. 3. An illustration to compute $VLBC_{1,8,1}$.

Based on the above discussions, we propose our idea and define the NR-VLBC as follows:

$$NR{-}VLBC_{L,P,R} = \min\{OVLBC_{L,P,R}, 3P+2-OVLBC_{L,P,R}\} \tag{28}$$

$$OVLBP_{L,P,R} = \sum_{p=0}^{3P+1} h(u_p - WLT) \tag{29}$$

$$h(u_p - WLT) = \begin{cases} 1, & \text{if } u_{p-1} - WLT > 0 \\ h(u_{p-1} - WLT), & \text{if } u_{p-1} - WLT = 0 \\ 0, & \text{if } u_{p-1} - WLT < 0 \end{cases} \tag{30}$$

where u_p is defined in Eq. (24). u_{-1} is set to u_{3P+1} by default.

The proposed NR-VLBC reflects the relative contrast between background and foreground, and so it is more robust and discriminative compared with the original VLBC. In addition, the original VLBC code and its complement are considered as the same case. Thus the dimension of feature vector will be reduced by half.

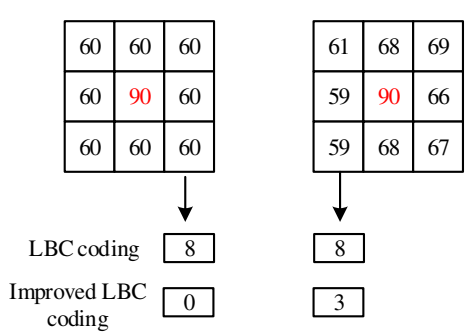

Fig. 4. Different patterns may have the same LBC code.

4.3.3. Completed features

We noted that the VLBC can model the local difference distribution, but it ignores the magnitude of gray-level differences and loses the local contrast information.

Based on the above considerations, the VLBC-Sign (VLBC_S), VLBC-Magnitude (VLBC_M), VLBC-Center gray level (VLBC_C) and VLBC-Difference (center symmetric) (VLBC_D) are introduced and combined to construct the Completed VLBC (CVLBC) descriptor. VLBC_S is equivalent to the proposed NR-VLBC, VLBC_M measures the local variance of magnitude. VLBC_C expresses the image local gray level, and also has discriminant information. VLBC_D expresses the gradient information of neighborhood pixels.

For convenience, we first introduce our ideas on the 2D texture descriptor. Fig. 6(a) is an original local structure with central pixel being 135. Fig. 6(b) is the difference vector (−3,−4,−3,0,2,8,2,5). Fig. 6(c) is the sign vector (−1,-1,-1,-1,1,1,1,1,1). Fig. 6(d) is the magnitude vector (3,4,3,0,2,8,2,5). Fig. 6(e) is the central symmetry difference vector (−5,−12,−5,−5). It is clearly seen that the original LBC uses only the sign vector to code the local pattern as an 8-bit string "00001111" ("−1" is coded as "0").

The VLBC_M can measure the local variance of magnitude, and it is defined as,

$$VLBC_M_{L,P,R} = \sum_{p=0}^{3P+1} s(m_p - c) \tag{31}$$

$$m_p = m_{x_c,y_c,t_c,p} = \left| g_{x_c,y_c,t_c,p} - WLT_{x_c,y_c,t_c,c} \right| \tag{32}$$

$$c = \frac{\sum\limits_{\substack{x_c = R+1,\ldots,W-R,\, y_c = R+1,\ldots,H-R,\\ t_c = L+1,..,T-L,\, p = 0,1,\ldots 3P+1.}} m_{x_c,y_c,t_c,p}}{(W-2R)(H-2R)(T-2L)(3P+2)} \tag{33}$$

where $g_{x_c,y_c,t_c,p}$ is the p th sample point of all the $3P+2$ ones. Threshold c is set as the mean value of $m_{x_c,y_c,t_c,p}$ of the whole image sequence. h

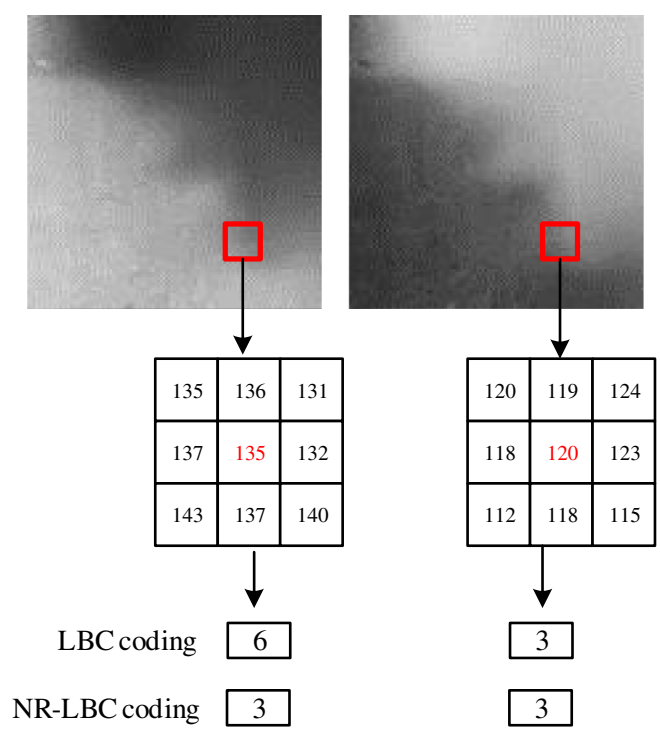

Fig. 5. Two images represent the same smoke structure with inverted background and foreground.

$(m_p - c)$ is defined in Eq. (30). W and H denote the block width and height, respectively. T denotes the frame number of the image sequence. R, P, L are the radius, the number of sample points and the time interval of the local volume, respectively.

The VLBC_S is equivalent to the proposed NR-VLBC in this paper and it is defined as

$$VLBC_S_{L,P,R} = NR{-}VLBC_{L,P,R} \tag{34}$$

where $NR - VLBC_{L,\,P,\,R}$ is defined in Eq. (28).

The VLBC_C has proven that the central pixel intensity can also provide useful discriminative information. The VLBC_C is defined as

$$VLBC_C_{L,P,R} = s(g_{t_c,c} - c_I) \tag{35}$$

where $s(x)$ and $g_{t_c,\,c}$ are defined in Eqs. (11) and (32). The threshold c_I is set as the average gray level of the whole image sequence.

The VLBC_D is defined as

$$VLBC_D_{L,P,R} = \sum_{p=0}^{3P/2} s(u_i - u_{i+3P/2+1}) \tag{36}$$

Given a local volume, we combine the four operators, including VLBC_S, VLBC_M, VLBC_C, and VLBC_D to form the Completed VLBC (CVLBC).

$$CVLBC = \{VLBC_S, VLBC_M, VLBC_C, VLBC_D\} \tag{37}$$

$$\begin{aligned} &VLBC_S \in \{0, 1, \ldots, 3P/2 + 1\}, \\ &VLBC_M \in \{0, 1, \ldots, 3P + 2\}, \\ &VLBC_C \in \{0, 1\}, \\ &VLBC_D \in \{0, 1, \ldots, 3P/2 + 1\} \end{aligned} \tag{38}$$

Two ways, including jointly or hybridly, can be used to combine the four operators. In this paper, we first build two 2-D joint histograms, "VLBC_S/D" and "VLBC_M/C". Then the histograms are converted to two 1D histograms. We concatenate the two 1D histograms to generate a joint histogram denoted by "VLBC_S/D_M/C",

The VLBC_S/D_M/C operator is given by

$$CVLBC_{S/D_M/C} = \{H_{S/D}, H_{M/C}\} \tag{39}$$

$$H_{S/D}(s, d) = \sum_{p=0}^{(3P/2+1)^2-1} f_1(CVLBC(p), s, d) \tag{40}$$

135	132	131
137	135	132
143	137	140

(a)

0	-3	-4
2		-3
8	2	5

(b)

-1	-1	-1
1		-1
1	1	1

(c)

0	3	4
2		3
8	2	5

(d)

-5	-5	-12
		-5

(e)

Fig. 6. (a) A local structure. (b) The local differences. (c) The sign. (d) The magnitude components. (e) The central symmetry difference.

$$f_1(R{-}VLBC, s, d) = \begin{cases} 1, & if\ \ VLBC_S(p) = s, \\ & VLBC_D(p) = d \\ 0, & otherwise \end{cases} \tag{41}$$

$$H_{M/C}(m, c) = \sum_{p=0}^{2(3P+2)-1} f_2(CVLBC(p), m, c) \tag{42}$$

$$f_2(R{-}VLBC, m, c) = \begin{cases} 1, & if\ \ VLBC_S(p) = m, \\ & \quad VLBC_D(p) = c \\ 0, & otherwise \end{cases} \tag{43}$$

where $H_{S/D}$ and $H_{M/C}$ are the two 2-D joint histogram of"VLBC_S/D" and "VLBC_M/C", respectively. The size of VLBC_S/D_M/C is $(3P/2 + 1)^2 + 2(3P + 2)$ bins, i.e., $(3P/2 + 1)(3P/2 + 4)$.

4.3.4. Multi-scale features

Because the smoke with different distances from the camera will present different scales, and so we propose to recognize the smoke in different scales.

Based on the above considerations, we introduce a multi-scale strategy into CVLBC to form the Robust VLBC (R-VLBC). There are mainly two ways to add multi-scale information. One is to change the R and P of the LBC operator, the other is down-sample original image with interpolation or low-pass filtering and then apply an LBC operator with fixed radius. In this paper, we obtain the multi-scale information by changing P and R.

Based on all the above discussions, the final feature vector of R-VLBC descriptor is given by

$$R{-}VLBC = \left\{CVLBC^{(s1)}_{S/D_M/C}, CVLBC^{(s2)}_{S/D_M/C}, CVLBC^{(s3)}_{S/D_M/C}\right\} \tag{44}$$

where $CVLBC^{(s)}_{S/D_M/C}$ denotes the $CVLBC_{S/D_M/C}$ with scale $s = (R,P) \in \{(1,8),(2,16),(3,24)\}$.

Compared with the original VLBC and existing other descriptors used in smoke detection, the proposed R-VLBC is more targeted and robust. The main characteristics include:

(1) The original VLBC is sensitive to the relative changes between the background and foreground. The proposed NR-VLBC, which is a part of R-VLBC, can solve this issue and reflect the relative contrast between the background and foreground. This makes the R-VLBC more robust and discriminative. In addition, the original VLBC code and its complement are considered as the same case to reduce intra-class distance. Thus the dimension of feature vector is reduced by half, and this is conducive to feature classification.
(2) The original VLBC is sensitive to noise since it only use the central pixel as a threshold directly. We use WLT as a threshold instead of the intensity value of the center pixel. Compared to the center pixel, WLT is more robust to noise and illumination variants. A parameter α is also used to make a balance between noise resistance and information of individual pixel.
(3) The VLBC_D is introduced to express the gradient information of neighborhood pixels. We combine VLBC_S, VLBC_M, VLBC_C and VLBC_D to form the CVLBC to characterize the spatial–temporal features. The VLBC_S is equivalent to the proposed NR-VLBC. The VLBC_M measures the local variance of magnitude. The VLBC_C expresses the image local gray level, and also has discriminant information.
(4) The smoke with different distances from the camera will present different scales, but the VLBC cannot express the scale information. The proposed R-VLBC can obtain multi-scale information by using a set of R and P to recognize smoke.

4.4. R-VLBC for smoke detection

We first extract suspected smoke blocks using the proposed R-Codebook model and partitioning strategy. For each suspected smoke

block, we extract its front block and back block in the same position of different frames to form a block sequence.

Then the R-VLBC features will be extracted from a block sequence and further feed to a trained support vector machine (SVM) classifier to classify the sequence to smoke or non-smoke.

The classification result of the block sequence will be taken as the classification result of the current block, which is the central block image of the block sequence. If the current block is classified as a smoke block, its corresponding frame will be classified as a smoke frame. In consecutive 100 frame, if more than N_{smoke} (8 is suggested) frames are classified as smoke frames, the current cropped video will be classified as a smoke video, i.e. a video with smoke vehicle passing by.

5. Experiments and analysis

In this section, extensive experiments will be conducted to verify the performances of the proposed method.

5.1. Datasets

Our data contains 102 traffic surveillance videos, including 98 short video and 4 long videos. Each short video has one smoke vehicle, and each long video has many non-smoke vehicle and 1~3 smoke vehicles. The videos totally have 5937 smoke frames and 151,613 non-smoke frames. All frames are downsampled to the size of 864 × 480 pixels. Some smoke frames and non-smoke frames can be seen in Fig. 7. The false alarms include vehicle shadow, tree shadow, shaking tree leaves and some smoke-liked vehicle body.

After extracting the suspected smoke blocks, four kinds of block image sequences are extracted to form dataset 1, dataset 2 dataset 3 and dataset 4, corresponding to smoke block sequences with heavy smoke vehicle, smoke block sequences with light smoke vehicle, non-smoke block sequences without shadow and non-smoke block sequences with shadow, respectively. It should be noted that these block sequences are extracted from consecutive frames of the video, and one block always occurs in more than one block sequences. Dataset 5 shows all the smoke frames and non-smoke frames. The datasets descriptions can be seen in Table 2.

5.2. Evaluation methods

The detection rate (DR) and false alarm rate (FAR) are used to evaluate the algorithm performances. DR is the probability that the smoke samples are correctly classified. FAR is the probability that non-smoke samples are wrongly classified as smoke samples. Their definitions are given by,

$$DR = \frac{TP}{P} \tag{45}$$

$$FAR = \frac{FN}{N} \tag{46}$$

where *TP and FN* are the numbers of true positive samples and false negatives, respectively. *P* and *N* are the number of all the positive samples and all the negative samples, respectively. The higher the DR as well as the lower the FAR, the better the method.

5.3. Performances of different methods on block sequences

To verify the advantages of the proposed method, some advanced smoke detection methods [3,18,25,40,6–9,12–15] are transplanted into smoke vehicle detection and used as baseline methods to make comparisons.

Table 3 shows the experimental results of different methods. We can see that our method achieves a better performance than all the other methods.

We noticed that all the methods achieve a higher DR on dataset 1 than dataset 2. This indicates that the light smoke block sequences are more difficult to be correctly recognized. In addition, all the methods achieve a lower FAR on dataset 3 than dataset 4. This indicates that the shadow is an important factor to cause false alarms.

Because the smoke block sequences on dataset 1 is heavy, most methods can achieve high DRs. It is difficult to further improve the DR greatly, and so our method only has a little higher DR than the other

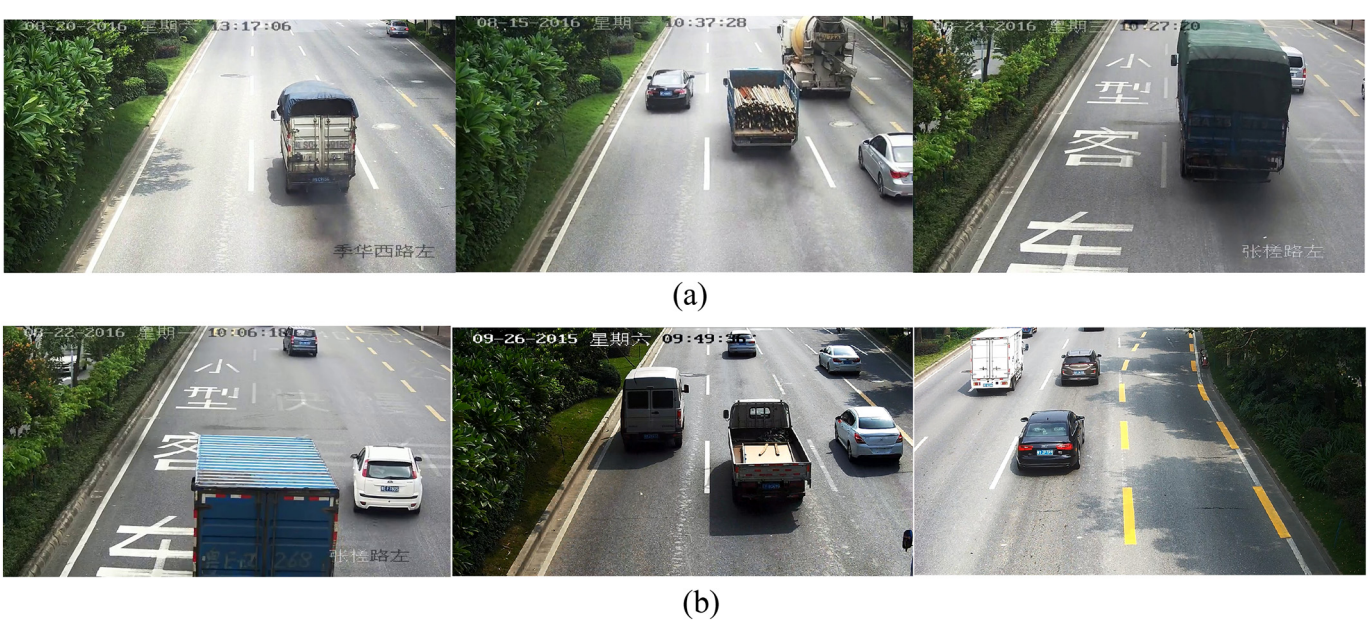

Fig. 7. Some smoke frames and non-smoke frames from the traffic surveillance videos. (a)Three smoke frames from the surveillance videos. (b) Three non-smoke frames from the surveillance videos.

Table 2
All the used datasets.

Datasets	Descriptions
Dataset 1	Contains 11,034 smoke block sequences with heavy smoke vehicles
Dataset 2	Contains 6618 smoke block sequences with light smoke vehicle
Dataset 3	Contains 1,341,890 non-smoke block sequences without shadows
Dataset 4	Contains 124,377 non-smoke block sequences with shadows
Dataset 5	Contains 5937 smoke frames and 151,613 non-smoke frames

methods. But on dataset 2, which is a more challenging dataset, our method achieves a much higher DR than the other methods. In addition, from the experimental results on dataset 3 and dataset 4, we can see that our method can reduce the false rates obviously, especially on dataset 4.

5.4. Performances of different methods on frame sequences

For real-world smoke vehicle detection in video, an alarm should be issued based on the whole frame instead of blocks. In our method, if the current block sequence is classified as a smoke block sequence, its corresponding central block will be classified as a smoke block. If there are one or more smoke block in the current frame, the current frame will be classified as smoke frame. Based on the classification results of block sequences, we can obtain the classification results of frames.

Table 3
Comparison experiments of different methods on block sequences classification.

Different methods	DR of smoke block sequences		FAR of non-smoke block sequences	
	Dataset 1	Dataset 2	Dataset 3	Dataset 4
Method in [3]	0.9603	0.7120	0.1520	0.2689
Method in [6]	0.9631	0.7135	0.1690	0.2985
Method in [7]	0.9634	0.7200	0.1844	0.2843
Method in [8]	0.9572	**0.7396**	0.1427	**0.2442**
Method in [40]	0.9514	0.6734	0.2344	0.3481
Method in [9]	0.9536	0.6884	0.2031	0.3035
Method in [12]	0.9581	0.6955	0.1879	0.2912
Method in [13]	0.9612	0.7013	0.1912	0.3044
Method in [14]	0.9588	0.7024	0.1887	0.2966
Method in [15]	0.9556	0.6993	0.1988	0.3132
Method in [18]	0.9577	0.7212	0.1436	0.2899
Method in [25]	**0.9667**	0.7243	**0.1320**	0.2961
Our method	**0.9733**	**0.8116**	**0.0930**	**0.1473**

Table 4
Experimental results of performance comparisons with other methods on frames classification.

Different methods	Dataset 5	
	DR	FAR
Method in [3]	0.8854	0.1331
Method in [6]	0.8958	0.1716
Method in [7]	0.8963	**0.1423**
Method in [8]	0.8701	0.1256
Method in [40]	0.8407	0.2301
Method in [9]	0.8452	0.1803
Method in [12]	0.8602	0.1610
Method in [13]	0.8871	0.1834
Method in [14]	0.8733	0.1703
Method in [15]	0.8550	0.2010
Method in [18]	0.8790	0.1433
Method in [25]	**0.9022**	0.1755
Our method	**0.9253**	**0.0893**

Table 5
Experiments results of using R-VLBC with different parameter α on block sequences classification.

R-VLBC with different α	DR of smoke block sequences		FAR of non-smoke block sequences	
	Dataset 1	Dataset 2	Dataset 3	Dataset 4
R-VLBC ($\alpha = 0$)	0.9617	0.8001	0.1044	0.1598
R-VLBC ($\alpha = 1/74$)	0.9710	0.8103	0.0950	0.1493
R-VLBC ($\alpha = 1/50$)	0.9733	**0.8116**	**0.0930**	0.1473
R-VLBC ($\alpha = 1/26$)	0.9735	0.8110	0.0968	**0.1456**
R-VLBC ($\alpha = 1$)	**0.9744**	0.8106	0.0972	0.1489

Table 4 shows the experimental results of different methods on frames classification. We can see that our method also achieves a better performance than the other methods.

If we give an alarm only based on that one frame is recognized as smoke frame, it is not unwise and will lead to false alarms. Therefore, we analyze the results of multiple frame sequences to make a decision. Such as, in consecutive 100 frame sequences, if more than N_{smoke} frame sequences are classified as smoke frame sequences, the current cropped video will be considered as a smoke video. By default, N_{smoke} is suggested to set to 8. To reduce false alarms, the user can increase N_{smoke}, but the detection rate will be reduced accordingly.

5.5. Discussions of the advantages of the proposed R-VLBC

In this part, we will investigate the proposed R-VLBC from the following four factors.

5.5.1. Discussion about noise resistance

To make the R-VLBC robust to noise and illumination variants, the WLT is introduced as a threshold instead of the intensity value of the center pixel. A parameter α is also used to make a balance between noise resistance and information of individual pixel.

The optimal parameter α in this paper is selected based on extensive experiments. Table 5 shows the experimental results of using R-VLBC with different parameter α. We can easily find that the R-VLBC ($\alpha = 1/50$) achieves a 1.15% higher DR than R-VLBC ($\alpha = 0$) on dataset 2, and achieves a 1.14% lower FAR than R-VLBC ($\alpha = 0$) on dataset 3. Thus, the R-VLBC ($\alpha = 1/50$) is more insensitive to noise than R-VLBC ($\alpha = 0$), while R-VLBC ($\alpha = 1/50$) performs more stably under complex illumination and viewpoint variant conditions, since it extracts the gray level information of both local neighbor set and individual pixel.

5.5.2. Discussion about non-redundant property

To reduce feature dimension and make the R-VLBC more robust and discriminative to reflect the relative contrast between the background and foreground. We propose the NR-VLBC, and take it as a term of R-VLBC.

Table 6 shows the experimental results of R-VLBC with and without non-redundant property. We can see the R-VLBC with non-redundant property achieves higher DR and lower FAR than the R-VLBC without non-redundant property. The reason is that adding non-redundant property is more robust and adaptive to relative changes of the background and foreground as well as achieves lower storage requirement.

Table 6
Experimental results of R-VLBC with and without non-redundant property on frames classification.

Different methods	Dataset 5	
	DR	FAR
R-VLBC without non-redundant property	0.9041	0.1082
R-VLBC with non-redundant property	**0.9253**	**0.0893**

Table 7
Experimental results of performance comparisons with other methods on frames classification.

Different methods	Dataset 5	
	DR	FAR
VLBC_S/D	0.9014	0.1211
VLBC_M/C	0.8957	0.1243
VLBC_S/D_M/C	**0.9253**	**0.0893**

Table 8
Experimental results of R-VLBC with different scales on frame sequences classification.

Different scales	Dataset 5	
	DR	FAR
R-VLBC with single scale 1	0.8997	0.1157
R-VLBC with single scales 2	0.9023	0.1097
R-VLBC with single scale 3	0.9012	0.1057
R-VLBC with multi scale	**0.9253**	**0.0893**

5.5.3. Discussion about competed property

We combine VLBC_S, VLBC_M, VLBC_C and VLBC_D to form the CVLBC to characterize the spatial–temporal features.

Table 7 shows the experimental results of VLBC_S/D_M/C VLBC_S/D, and VLBC_M/C. We can see that the VLBC_S/D_M/C achieves a better performance than the other descriptors.

5.5.4. Discussion about multi-scale strategy

We introduce a multi-scale strategy into the R-VLBC to recognize smoke by using a set of R and P to extract multi-scale information.

Table 8 shows the experimental results of R-VLBC with single scale and multiple scales. We can see that the R-VLBC with multiple scales achieves a better performance than the single scale.

Table 9
Experiments to verify the advantages of R-Codebook model on frame classification.

Different methods	Dataset 5	
	DR	FAR
Original Codebook model	0.9094	0.1153
R-Codebook model	**0.9253**	**0.0893**

5.6. Experimental comparison to verify the performance of R-Codebook model

In Section 2, we propose the R-Codebook model to detect suspected smoke blocks. It is robust to illumination changes in the scene as well as the camera shaking and leaf shaking. To verify the advantages, we keep other algorithm steps unchanged, and replace R-Codebook by the original Codebook model. We evaluate the algorithm robustness through the recognition results, since the purpose of the improvement is to improve the detection rate and reduce the false alarm rates.

Table 9 shows the experimental results. We can see that the DR of R-Codebook model is higher than that of the original Codebook model. This verifies that our improvements are necessary. From the further analysis of the wrong classified samples, we find that, the R-Codebook model eliminates the detected moving object caused by leaf shaking and camera shaking, and so it is more robust and can achieve lower false alarm rates compared with the original Codebook model.

5.7. Some smoke frames captured from surveillance video

Two smoke frames captured from traffic surveillance videos can be seen in Fig. 8. The test results show the robustness and effectiveness of the proposed method.

6. Discussion and conclusion

This paper presents a smoke vehicle detection method based on R-Codebook Model and R-VLBC to reduce false alarm rates of smoke vehicles. The R-Codebook model is proposed to detect suspected smoke blocks, and the R-VLBC is proposed to recognize the smoke block sequence. Firstly, to overcome the issue that the original VLBC is sensitive to the relative changes between the background and foreground, we propose the NR-VLBC, which is robust and discriminative with half feature dimension. Secondly, we propose the CVLBC by combining VLBC_S, VLBC_M, VLBC_C and VLBC_D to characterize the spatial–temporal features. Thirdly, we further improve the robustness of CVLBC to propose R-VLBC based on two strategies. One is to overcome the issue that the original LBC is sensitive to noise based on the way to replace the threshold of the central pixel by the WLT, which is robust to noise and illumination variants and make a balance between noise resistance and information of individual pixel. The other one is to extract multi-scale information by using a set of radius and sampling points to recognize smoke block sequences.

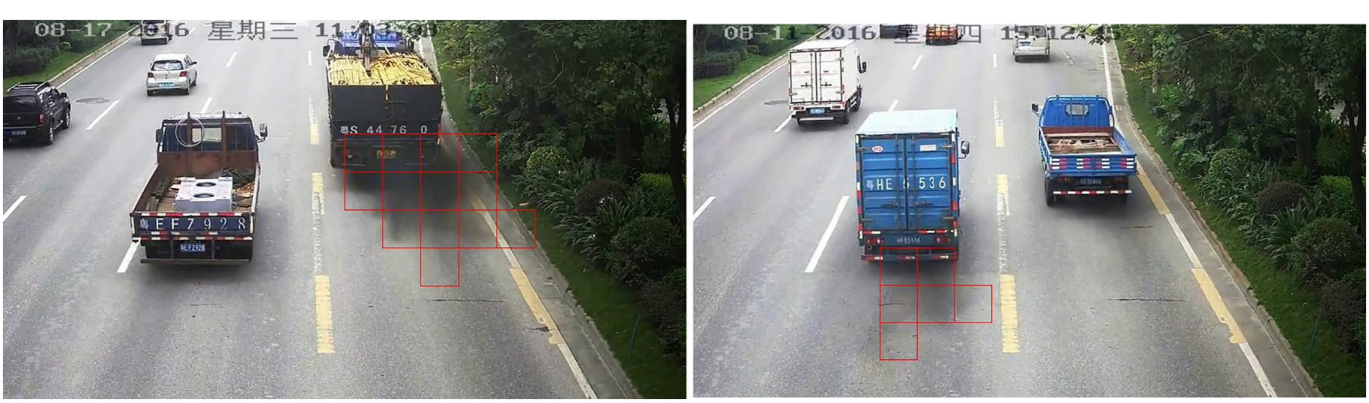

Fig. 8. Some smoke frames captured from surveillance videos. The red blocks denote the correctly recognized smoke blocks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

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